

OptiCrop

Smart Crop Recommendation System

Full Stack Project Documentation

Team Member	Role
Bala Sai Venkat Ramisetty	Team Lead
Eede Veda Srivalli	Member
Marneni Pranathi	Member
M Anunay	Member
Kadingu Uday Kumar	Member

1. Introduction

Project Title

OptiCrop – Smart Crop Recommendation System

Team Members

Bala Sai Venkat Ramisetty — Team Lead

Eede Veda Srivalli — Member

Marneni Pranathi — Member

M Anunay — Member

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2. Project Overview

Purpose

OptiCrop is a web-based smart agriculture application that recommends a suitable crop from soil nutrient values and environmental conditions. The goal is to convert seven simple inputs—Nitrogen, Phosphorus, Potassium, Temperature, Humidity, Soil pH and Rainfall—into an understandable crop recommendation supported by crop details, ideal conditions and growing guidance.

Features

- Seven-parameter crop recommendation form
- Machine-learning-based crop prediction
- Recommended crop result card
- Crop season, harvest time, soil type and water requirement information
- Fertilizer, market demand, suitable region and common-use guidance
- Ideal environmental condition display
- Growing tips
- “How OptiCrop Works” workflow explanation
- Model information: 2200+ training records, 22 crop classes, 7 input features and Random Forest algorithm
- Responsive, section-based web interface

3. Architecture

Frontend

The user interface is a responsive single-page style website built with HTML, CSS and JavaScript. Navigation sections include Home, Predict, Crop Details, How It Works and Dataset/Model information. JavaScript handles user interaction and submission of prediction values.

Backend

The application uses a Python Flask backend. The server receives the seven crop parameters, validates and preprocesses them, sends them to the trained machine-learning model, and returns the predicted crop for display.

Machine Learning Layer

A Random Forest classification model is used for crop recommendation. The model uses seven features: N, P, K, temperature, humidity, pH and rainfall. The website reports 2200+ training records and 22 crop classes.

Database / Data Layer

The demonstrated version is prediction-oriented and does not show user-account persistence. The crop dataset is used for model training, while crop descriptive information is served to the result interface. A conventional database can be added later for users, prediction history, feedback and analytics.

4. Setup Instructions

Prerequisites

Python 3.x, pip, Flask, pandas, NumPy, scikit-learn, joblib or pickle, and a modern web browser.

Installation

1. Copy or clone the project source code.
2. Create and activate a Python virtual environment.
3. Install dependencies using the project requirements file.
4. Place the trained model and required dataset/resources in their configured folders.
5. Configure Flask environment settings if required.
6. Run the Flask application and open the local address shown in the terminal.

5. Folder Structure

Suggested Organization

```
OptiCrop/  
■■■■ app.py  
■■■■ model/  
■ ■■■■ crop_recommendation_model.pkl  
■■■■ templates/  
■ ■■■■ index.html  
■■■■ static/  
■ ■■■■ css/  
■ ■■■■ js/  
■ ■■■■ images/  
■■■■ data/  
■ ■■■■ crop_recommendation.csv  
■■■■ requirements.txt
```

6. Running the Application

Local Run

Activate the virtual environment, install dependencies, then run the Flask entry point (for example, `python app.py` or `flask run` depending on project configuration). Open the local URL displayed by Flask in a browser and use the Crop Recommendation form.

7. API Documentation

Prediction Endpoint

Typical application flow: POST `/predict`. Request fields: nitrogen, phosphorus, potassium, temperature, humidity, ph and rainfall. The backend converts values to the model input format and returns the recommended crop. Example logical response: `{"recommended_crop": "rice"}`. Exact route naming may vary with the source implementation.

Page Endpoint

GET `/` returns the main OptiCrop interface.

8. Authentication

Current Scope

The demonstrated application does not show login, registration, token or session-based authorization. The crop recommendation feature is publicly accessible. If user profiles and saved prediction history are added, secure

password hashing, session management and role-based authorization should be implemented.

9. User Interface

UI Description

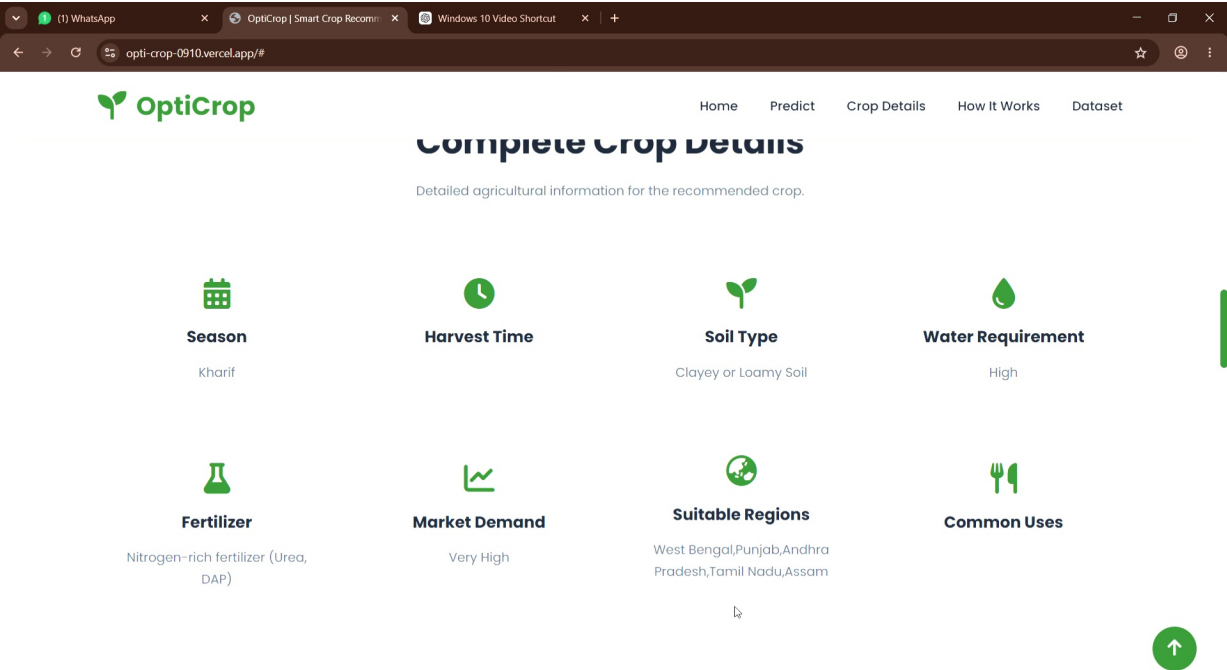
The interface uses a clean green agricultural theme. The home area introduces smart farming, the prediction form collects seven values, the result area shows the recommended crop, and later sections present crop details, environmental conditions, growing tips, workflow and model information.

The screenshot shows the OptiCrop web application interface. The browser tabs include WhatsApp, OptiCrop | Smart Crop Recommendation, and Windows 10 Video Shortcut. The address bar shows the URL `opti-crop-0910.vercel.app/#`. The navigation bar contains links for Home, Predict, Crop Details, How It Works, and Dataset. The main content area features a green header with the OptiCrop logo and a navigation bar. Below the header, there is a section titled "Smart Agriculture" with the text "Smart Farming Starts Here". A sub-header "Crop Recommendation" is followed by the instruction "Enter the soil and environmental values." The input form consists of five rows, each with a label and a text input field: Nitrogen (N) with value 60, Phosphorus (P) with value 40, Potassium (K) with value 32, Temperature (°C) with value 32, and Humidity (%) with value 85. At the bottom, there are three statistics: 22 Crop Classes, 2200+ Records, and ML Prediction.

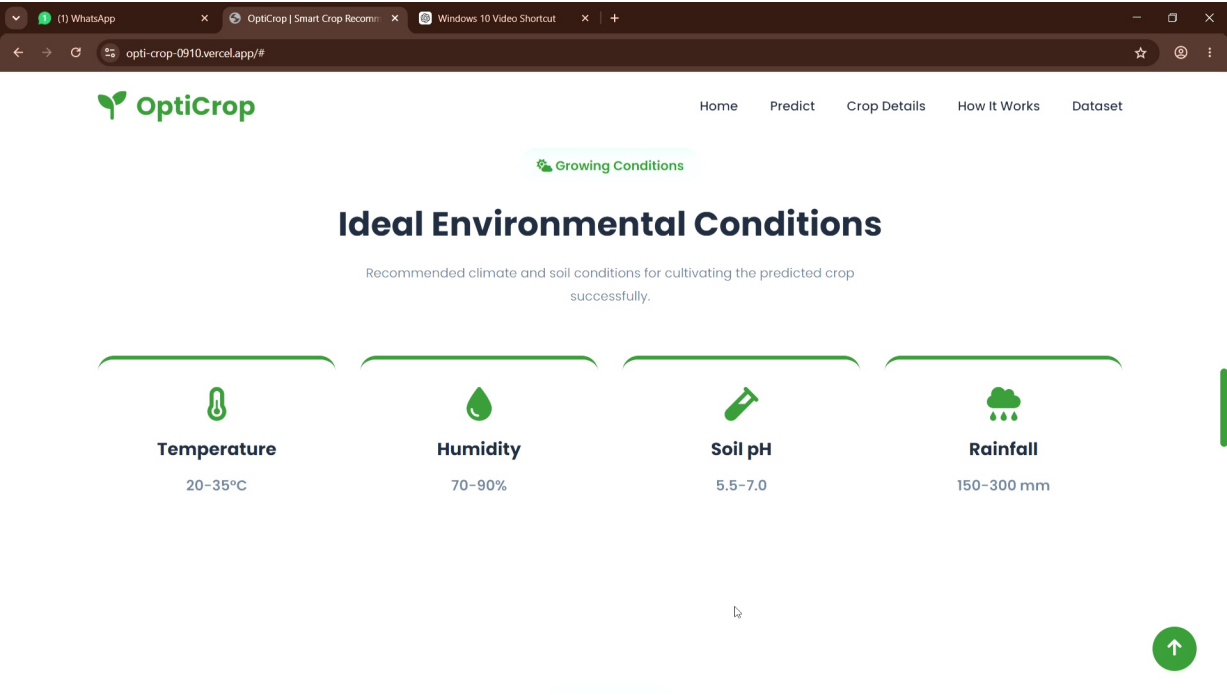
Home and crop input interface

The screenshot shows the OptiCrop web application interface displaying the prediction result. The browser tabs and address bar are the same as in the previous screenshot. The navigation bar remains the same. The main content area features a green header with the OptiCrop logo and a navigation bar. Below the header, there is a section titled "Prediction Result" with the text "Recommended Crop". A sub-header "Best Recommendation" is followed by the text "Rice". Below the text, there is a description of rice: "Rice is one of the world's most important staple food crops. It thrives in warm and humid climates with abundant water and is widely cultivated across Asia." To the left of the text is a photograph of a rice field. Below the text, there is a green box labeled "Prediction Confidence" with the value "52 %". A green circular button with an upward arrow is located at the bottom right.

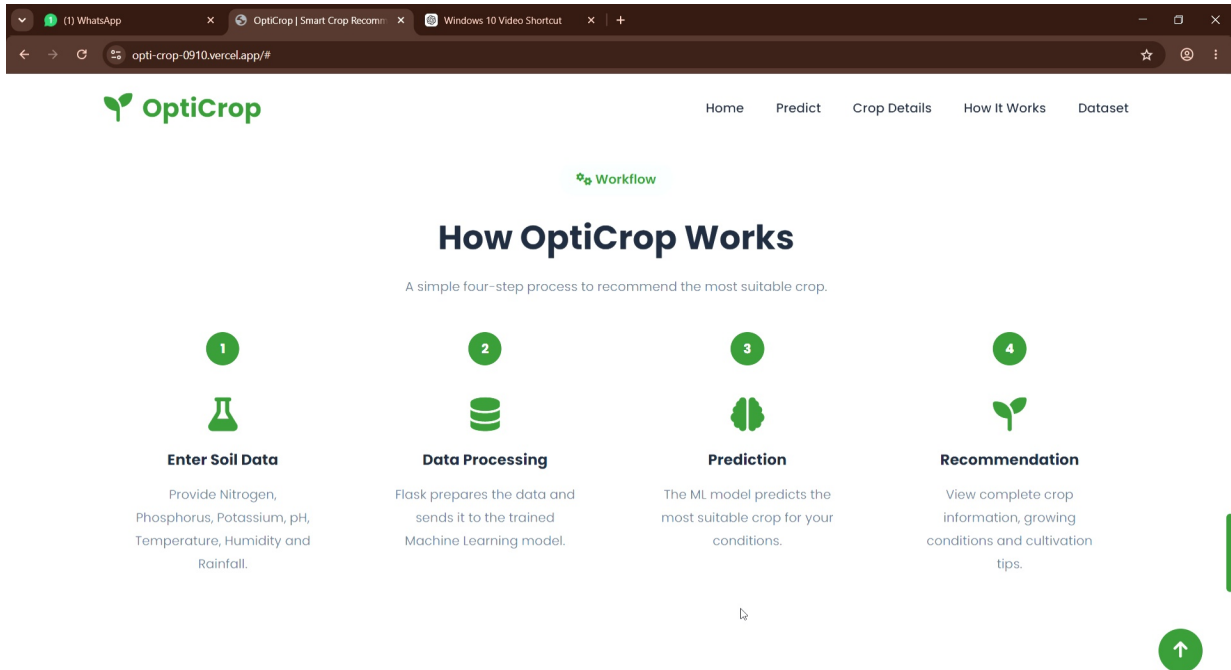
Recommended crop result



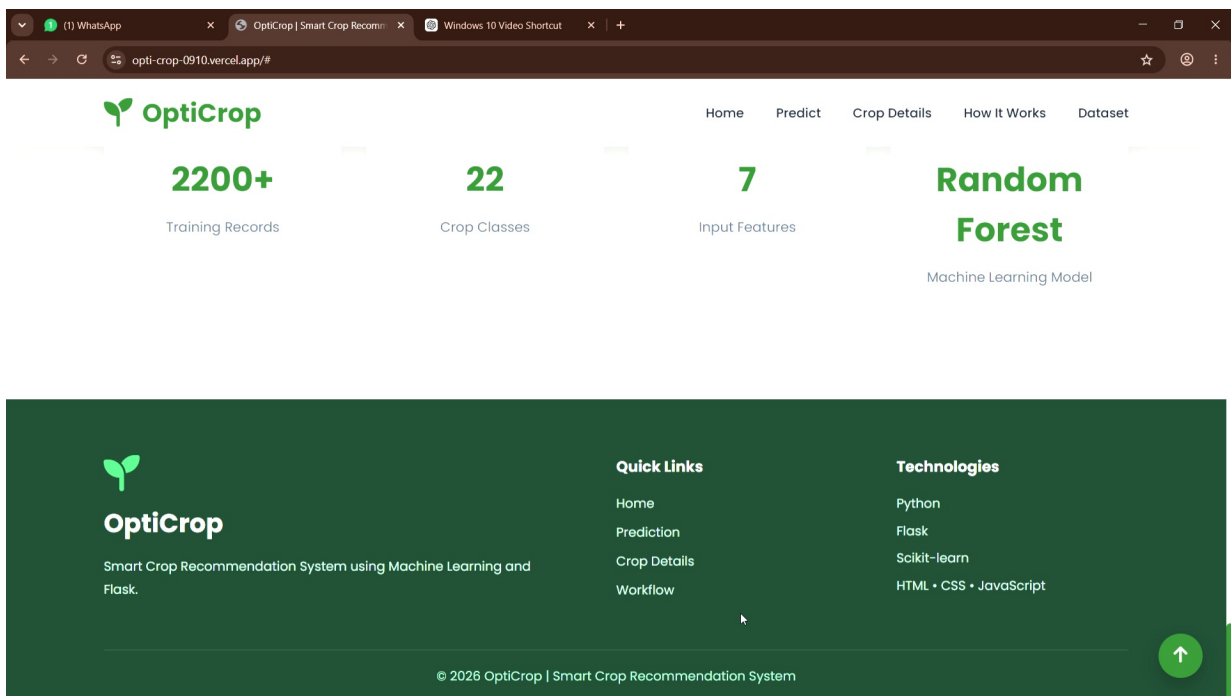
Crop details



Ideal environmental conditions



System workflow



Model information

10. Testing

Strategy

Testing should cover input validation, prediction response, route handling, model loading, invalid or missing values, boundary values and browser responsiveness. Functional tests verify that all seven fields can be submitted and a crop result is displayed. Model tests compare predictions with held-out labelled data. UI tests verify navigation, layout and readability on common screen sizes.

11. Screenshots or Demo

Demo Coverage

The uploaded project demo shows the home and prediction interface, a rice recommendation result, crop details, ideal environmental conditions, growing tips, the four-step OptiCrop workflow and model information.

12. Known Issues

Limitations

- Recommendation quality depends on the representativeness and correctness of training data.
- Incorrect sensor or manually entered values can produce unsuitable recommendations.
- The current demo does not show location-aware weather integration, live market prices or user history.
- The model output should support—not replace—local agronomic expertise, soil testing and regional agricultural advisories.

13. Future Enhancements

Planned Improvements

Integrate live weather APIs, GPS/location-aware recommendations, soil sensor or IoT input, multilingual support, fertilizer recommendation, pest and disease alerts, crop yield estimation, market-price forecasting, user accounts with prediction history, explainable-AI confidence information and mobile/PWA support.